

Payroll & Workforce Analysis

Project Overview

This project presents an end-to-end **payroll and workforce analysis** using employee and salary data from an organizational payroll workbook.

The objective was to transform raw payroll data into actionable business insights that could support **management decision-making, workforce planning, compensation review, payroll control, and budgeting**.

The analysis follows a professional data analytics workflow:

Data Inspection → Data Cleaning → Validation → Exploratory Analysis → KPI Development → Visualization → Insights → Recommendations

Business Objectives

The analysis was designed to answer the following questions:

- How many employees are in the organization?
 - What is the total payroll cost?
 - Which departments account for the largest share of payroll?
 - How are employee salaries distributed?
 - Are there notable differences in salary between employee groups?
 - How are gross salary, deductions and net salary calculated?
 - Are the payroll calculations internally consistent?
 - Are there data-quality or reporting-control issues?
 - What actions should management consider?
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Dataset

The source workbook contains employee and payroll information including:

- Employee ID
- Gender

- Age
- Department
- State
- Hire Date
- Basic Salary
- Performance Bonus
- House Allowance
- Transport Allowance
- Wardrobe Allowance
- Overtime
- Gross Salary
- Loan Deduction
- Tax
- Health Insurance
- Retirement Plan
- Total Deductions
- Net Salary

The analysis covered **50 employees across 8 departments**.

 Key Findings

Payroll Overview

KPI	Result
Employees	50
Monthly Basic Payroll	₦17.22M
Monthly Gross Payroll	₦21.525M

KPI	Result
Monthly Deductions	₦1.722M
Monthly Net Payroll	₦19.803M
Average Basic Salary	₦344,400
Median Basic Salary	₦327,500
Lowest Basic Salary	₦180,000
Highest Basic Salary	₦600,000

The workbook's salary model increases basic salary by **25% through allowances** and applies deductions equivalent to **10% of basic salary**.

Department Analysis

The analysis found that **ICT is the largest payroll cost centre**, accounting for approximately **21.4% of basic payroll**.

Department-level analysis included:

- Headcount
- Total basic payroll
- Gross payroll
- Total deductions
- Net payroll
- Average basic salary
- Average net salary
- Payroll share

This provides management with a clearer understanding of where compensation costs are concentrated.

Salary Distribution

Employees were grouped into four salary bands:

- ≤ ~~₦~~250K
- ~~₦~~250K–~~₦~~350K
- ~~₦~~350K–~~₦~~450K
- ~~₦~~450K

A key finding was that **24 of the 50 employees (48%)** fall within the ~~₦~~250K–~~₦~~350K basic salary range.

This suggests that employee compensation is relatively concentrated within the middle of the salary distribution.

Workforce & Pay Equity Analysis

The workforce is evenly split by gender, with:

- 25 female employees
- 25 male employees

Average basic salary:

Gender Average Basic Salary

Male ~~₦~~384,400

Female ~~₦~~304,400

This represents an **unadjusted average salary difference of approximately 20.8%**.

Importantly, this should not automatically be interpreted as evidence of discriminatory pay.

A proper pay-equity analysis would additionally control for:

- Job role
- Job grade
- Department
- Seniority
- Tenure

- Experience
- Management level

The result is therefore treated as a **signal requiring further investigation**, rather than a definitive conclusion.

Workforce Experience

The workforce has an average tenure of approximately **8.5 years** as of September 2026.

This indicates a relatively experienced workforce and raises opportunities for further analysis around:

- Employee retention
 - Career progression
 - Succession planning
 - Salary progression
 - Long-service compensation
 - Workforce planning
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Payroll Calculation Validation

The payroll formulas were independently checked.

The primary salary data sheet follows these relationships:

Gross Salary = Basic Salary × 1.25

Total Deductions = Basic Salary × 0.10

Net Salary = Basic Salary × 1.15

The component allowances are:

Performance Bonus = 1%

House Allowance = 15%

Transport Allowance = 5%

Wardrobe Allowance = 2%

Overtime = 2%

Total allowances:

25% of Basic Salary

Deductions:

Loan = 1%

Tax = 5%

Health Insurance = 2%

Retirement Plan = 2%

Total deductions:

10% of Basic Salary

Data Quality Finding

One of the most important findings was a discrepancy between worksheets.

The salary data sheet contains internally consistent calculations, while the displayed values on the DATA IN A RANGE sheet do not fully reflect the same calculations.

For example, the displayed gross payroll on DATA IN A RANGE was approximately:

₦17.22M

compared with:

₦21.525M

from the validated salary data calculations.

This appears consistent with a **stale/unrecalculated Excel formula cache** rather than a different payroll methodology.

Recommended control

Before the workbook is used for official payroll reporting:

1. Recalculate all workbook formulas.
 2. Reconcile totals between worksheets.
 3. Establish a single source of truth.
 4. Remove or clearly label duplicate reporting sheets.
 5. Introduce automated payroll validation checks.
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Analysis & Visualisations

The project includes analysis of:

- Payroll by department
 - Salary distribution
 - Average salary by gender
 - Workforce demographics
 - Hiring patterns
 - Payroll composition
 - Payroll calculation structure
 - Data-quality controls
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Business Recommendations

1. Establish a single payroll source of truth

Eliminate conflicting versions of payroll calculations and clearly identify the authoritative dataset.

2. Strengthen payroll controls

Implement checks for:

- Duplicate employee IDs
- Missing information

- Formula errors
- Incorrect deductions
- Gross/net salary reconciliation

3. Review compensation architecture

Introduce formal:

- Job grades
- Salary bands
- Role-based compensation
- Career progression frameworks

4. Conduct a detailed pay-equity review

Investigate the observed gender salary difference using like-for-like comparisons.

5. Improve management reporting

Develop a recurring payroll dashboard covering:

- Headcount
- Basic payroll
- Gross payroll
- Net payroll
- Department payroll
- Average salary
- Salary distribution
- Payroll trends

Tools & Skills Demonstrated

Tools

- Microsoft Excel
- Data analysis

- Data validation
- Data visualisation
- Report development

Analytical Skills

- Data cleaning
- Exploratory Data Analysis (EDA)
- KPI development
- Aggregation and segmentation
- Salary distribution analysis
- Workforce analysis
- Data-quality assessment
- Business storytelling
- Stakeholder reporting
- Insight generation
- Recommendation development

Project Deliverables

The project contains:

Payroll & Workforce Analysis

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| — Source Data

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| — Payroll workbook

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| — Analysis

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| — Payroll KPIs

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| — Department analysis

- | | — Salary distribution
 - | | — Workforce analysis
 - | | — Data-quality validation
 - |
 - | — Visualisations
 - | | — Payroll by Department
 - | | — Salary Distribution
 - | | — Average Salary by Gender
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 - | — Reports
 - | — Stakeholder Analysis Report
 - | — Supporting Excel Analysis
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Key Takeaway

This project demonstrates how raw payroll data can be transformed into a structured management report that goes beyond reporting numbers.

The analytical process connects:

Data → Validation → Analysis → Insight → Business Impact → Recommendation

The project also demonstrates an important principle in professional analytics:

A good analyst does not only identify insights; they also challenge the quality and reliability of the data behind those insights.

Author

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Data Analytics | Business Intelligence | Excel | Data Visualisation
